

Football Analytics Blitz



Optimizing Kickoff Strategy
and Injury Prevention



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Contents

KICKOFF STRATEGIES

1 YEAR OVER YEAR CHANGE

2 2024 ANALYSIS

3 STATISTICAL ANALYSIS

INJURY PREVENTION

4 YEAR OVER YEAR CHANGE

5 STATISTICAL ANALYSIS

6 PREDICTIVE MODELING

7 FIELD SURFACES

CONCLUSION

8 OUR ADVICE

9 Q&A

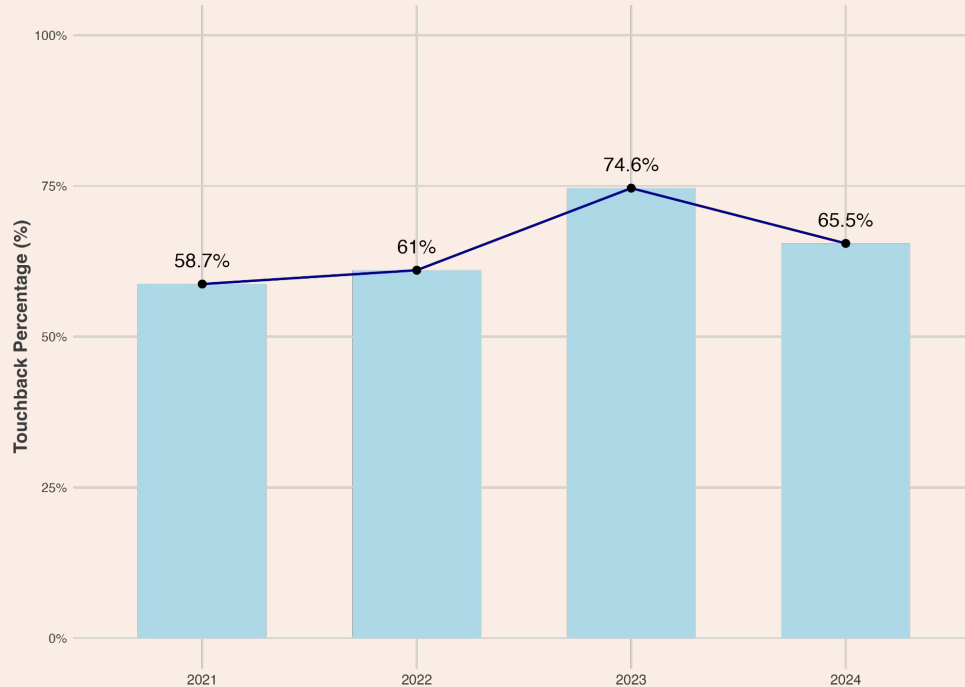
Kickoff Strategies





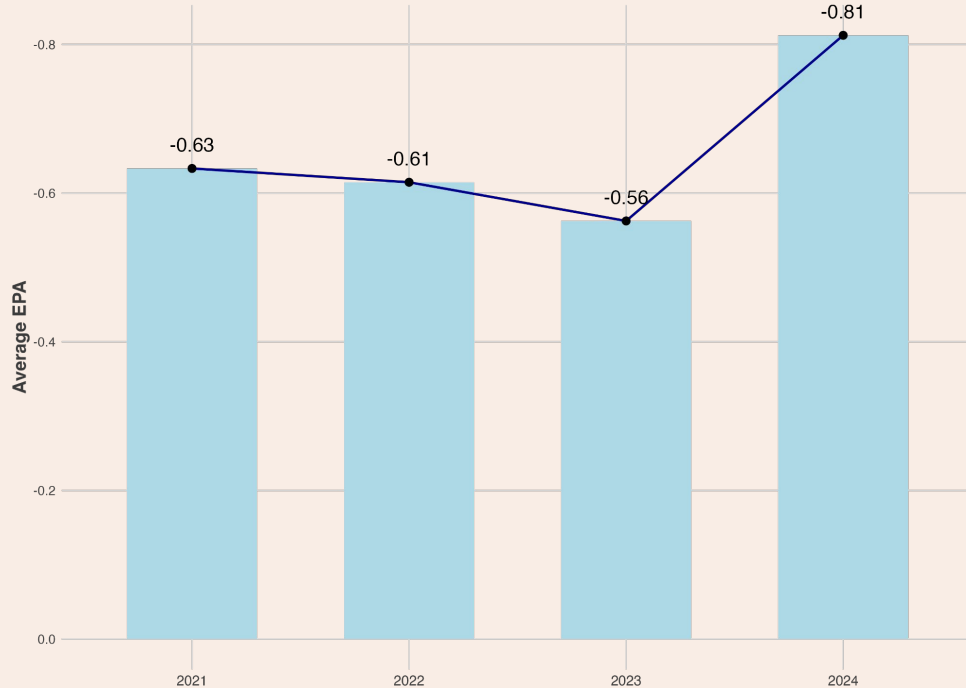
Year over Year Change

Touchback Percentage by Season



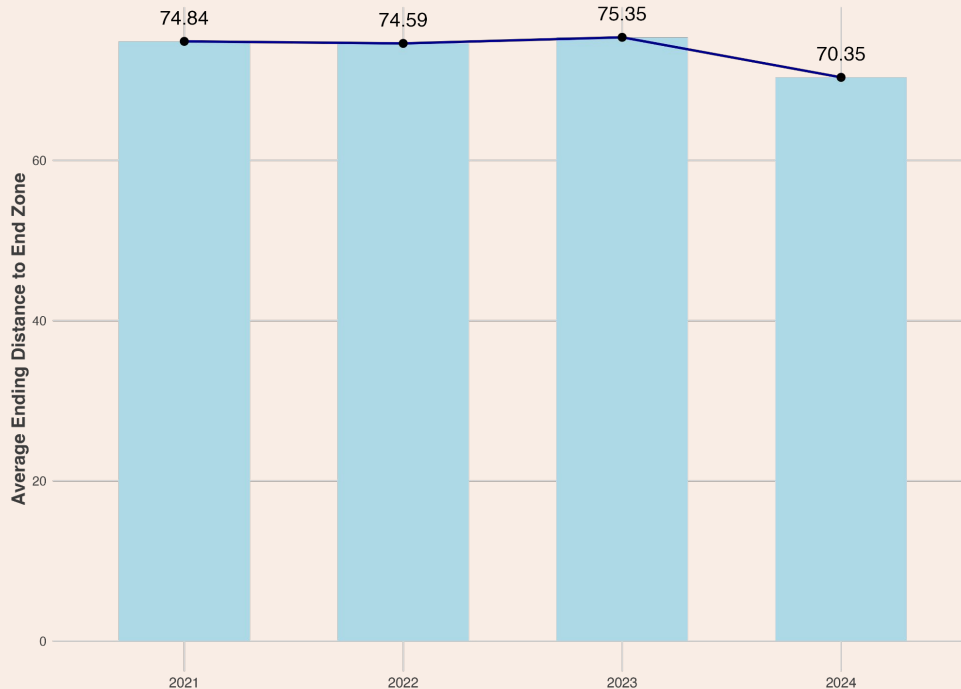
Dynamic Kickoff brought touchback percentage back to the 60% range from ~75%

Expected Points Added (EPA) Distribution by Season



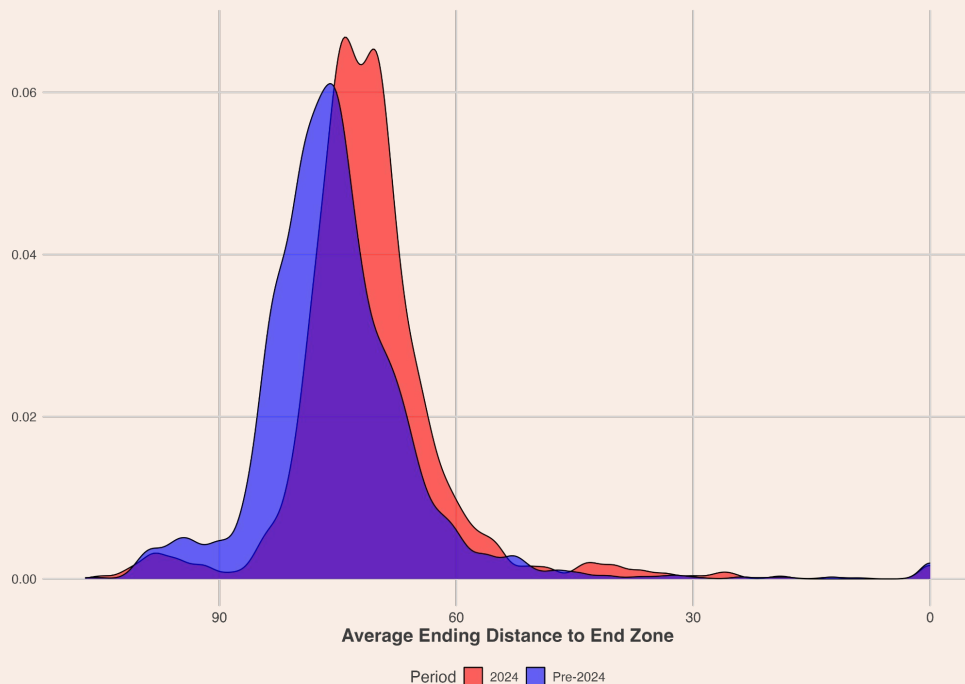
The kicking team lost more expected points in 2024 than the three years before.

Average Ending Distance to End Zone Distribution by Season



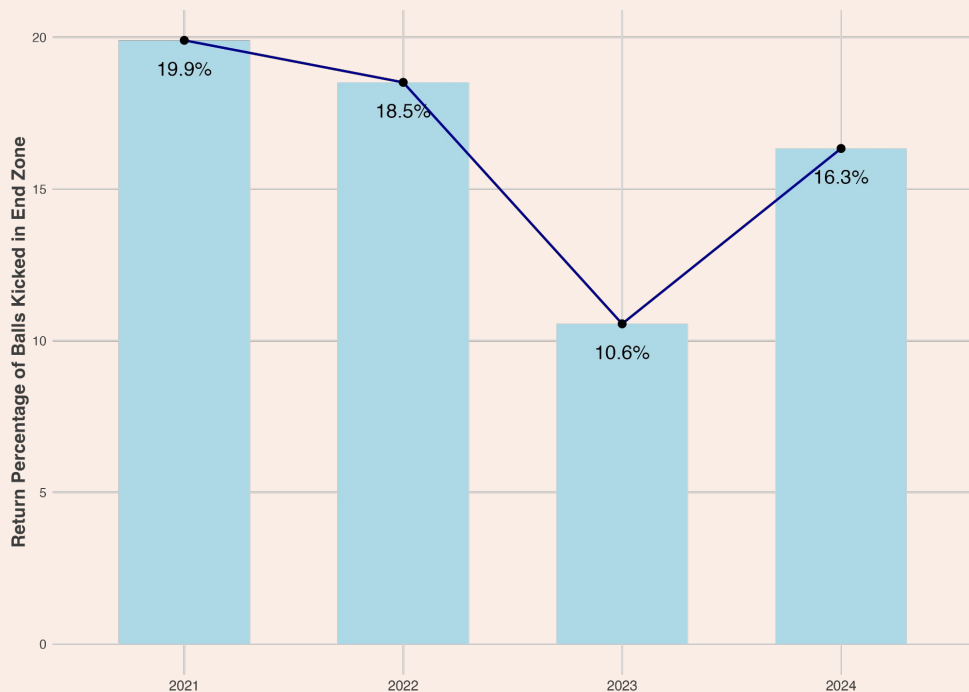
There is a 5 yard difference in 2024 than years prior, mainly due to new touchbacks rules.

Density of Average Ending Distance to End Zone (Pre-2024 vs 2024)



Clear skew right due to new touchback rules.

Return Percentage of Kicks in End Zone Distribution by Season

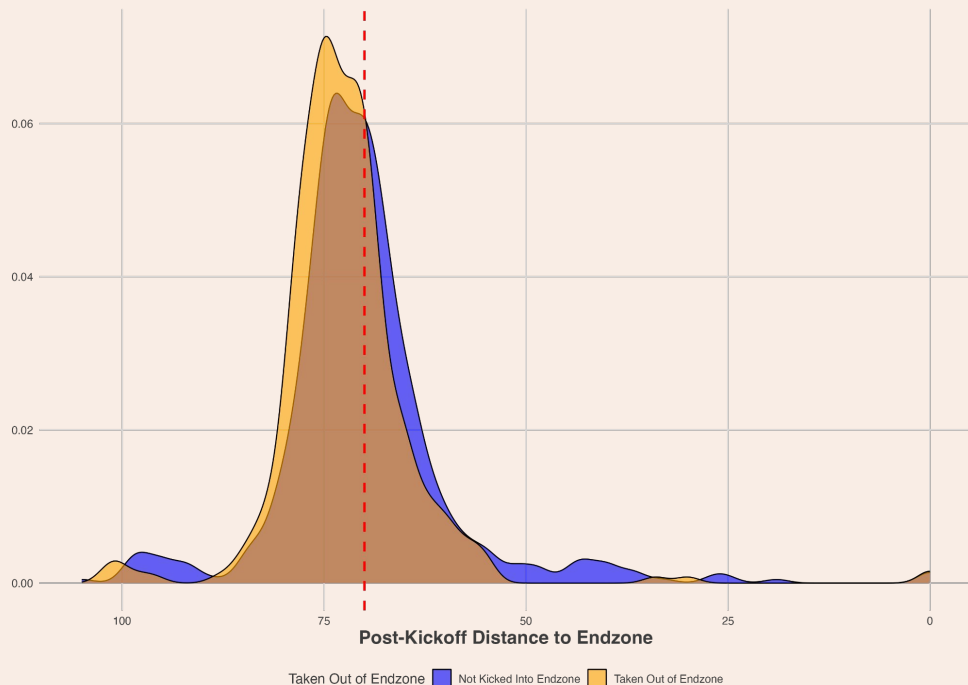


Return percentage increased 53.8% from 2023 to 2024.



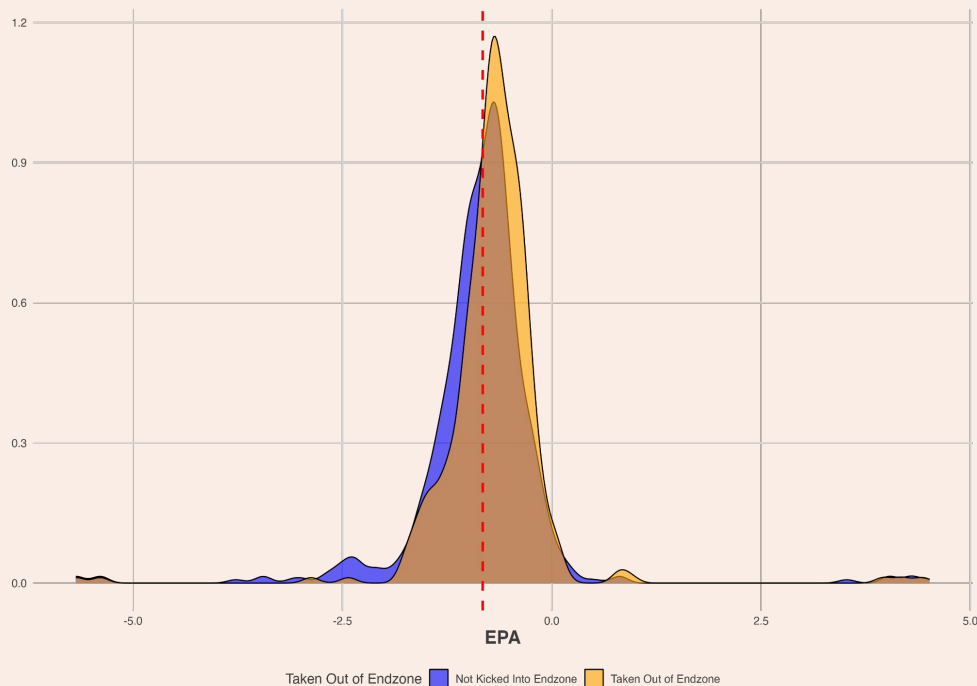
2024 Analysis

Density of Post-Kickoff Distance to Endzone When Returned



Returners do worse when taking it out of the end zone than from landing zone.

Density of EPA (No Touchback)



Clear EPA loss when the returner takes the ball out of the end zone.

Analyzing Success of Kickoff Decisions

- 1 It is great for kicking team when players return the ball from the end zone in the first 3 quarters.
- 2 When teams kick it to the landing zone and are losing, it hurts the kicking team more than if they were winning.
- 3 When kicked for a touchback, EPAs are all very similar, as you would expect given a set post-kickoff yardline.
- 4 This table shows what you can expect given circumstances in a game for each kick decision.

Outcomes of Kickoff Strategies (2024)

Game State of Kicking Team	Avg Kicking Team EPA	Avg Distance to End Zone
Returned from End Zone - First 3 Quarters		
Winning	-0.64	71.92
Losing	-0.65	73.80
Returned from End Zone - 4th Quarter and OT		
Winning	-0.65	72.96
Tying	-0.84	70.38
Losing	-0.76	71.46
Returned from Landing Zone - First 3 Quarters		
Winning	-0.83	69.95
Losing	-1.04	70.17
Returned from Landing Zone - 4th Quarter and OT		
Winning	-0.80	70.19
Tying	-0.93	69.74
Losing	-0.84	69.81
Touchback - First 3 Quarters		
Winning	-0.81	70.28
Tying	-0.68	70.22
Losing	-0.84	70.18
Touchback - 4th Quarter and OT		
Winning	-0.83	70.27
Tying	-0.86	70.30
Losing	-0.80	70.32
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Analyzing Success of Kicking Into End Zone vs. Landing Zone

- 1 In the first 3 quarters, it is always better to kick in the end zone rather than the landing zone, regardless of score.
- 2 When the kicking team is winning late, there is a minimal difference between kicking location.
- 3 Returning from the landing zone is slightly better for the return team in terms of distance to the end zone.
- 4 Making decisions based on the score is only recommended if the kicking team is losing in the first 3 quarters.

Outcomes of Kickoff Strategies (2024)

Game State of Kicking Team Avg Kicking Team EPA Avg Distance to End Zone

First 3 Quarters - Kicked in End Zone

Winning	-0.78	70.54
Tying	-0.67	70.67
Losing	-0.82	70.61

First 3 Quarters - Kicked in Landing Zone

Winning	-0.83	69.95
Tying	-0.88	68.80
Losing	-1.04	70.17

4th Quarter and OT - Kicked in End Zone

Winning	-0.80	70.72
Tying	-0.85	70.31
Losing	-0.79	70.54

4th Quarter and OT - Kicked in Landing Zone

Winning	-0.80	70.19
Tying	-0.93	69.74
Losing	-0.84	69.81

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Statistical Analysis

Statistical Significance

After running various statistical tests, such as a t-test and an asymptotic two sample Kolmogorov-Smirnov test, we can determine that there IS a statistical significance with the overall distribution of returns.



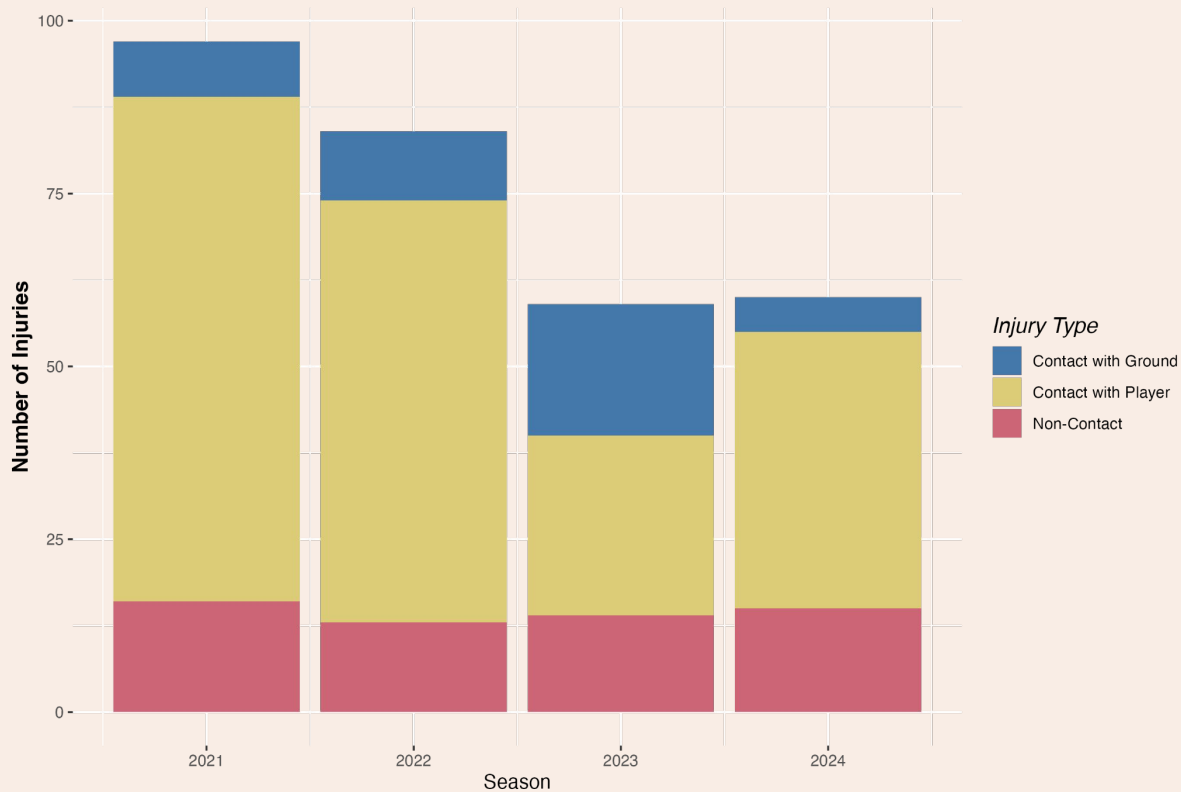
Injury Prevention





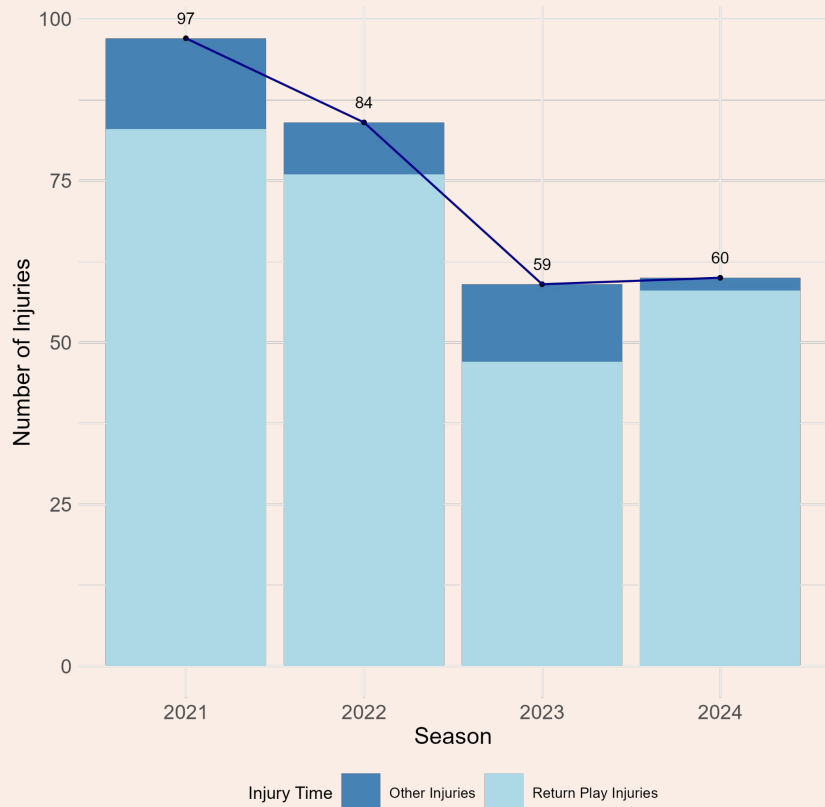
Year Over Year Change

Contact and Non-Contact Injuries by Season



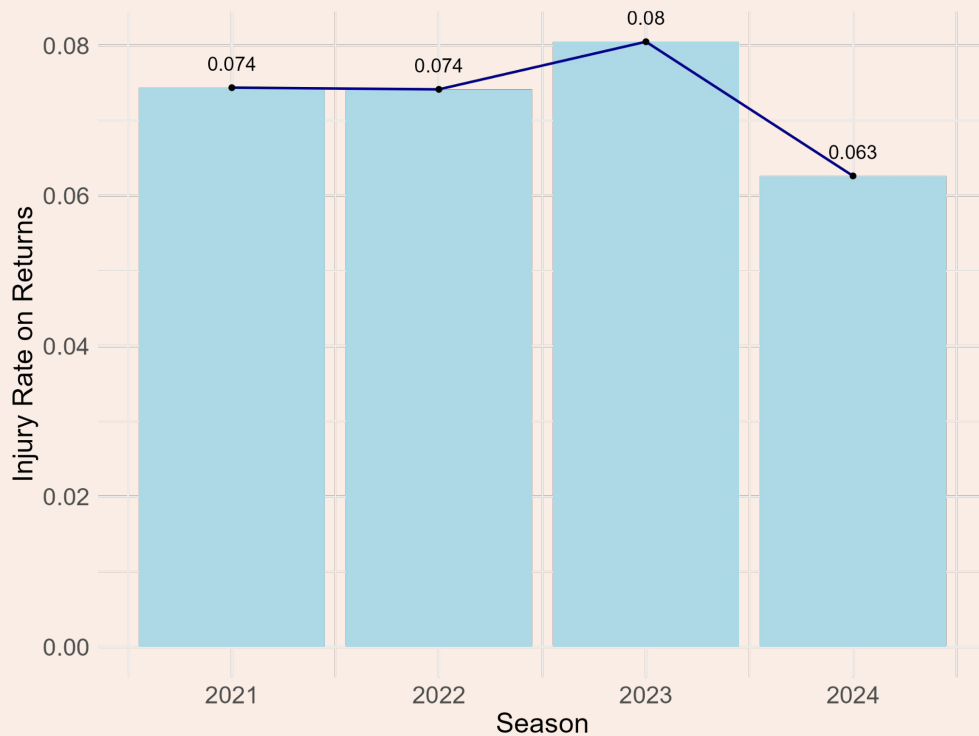
Player to player injuries increased, where player to ground injuries decreased greatly.

Injuries on Returned and Non-Returned Kickoffs



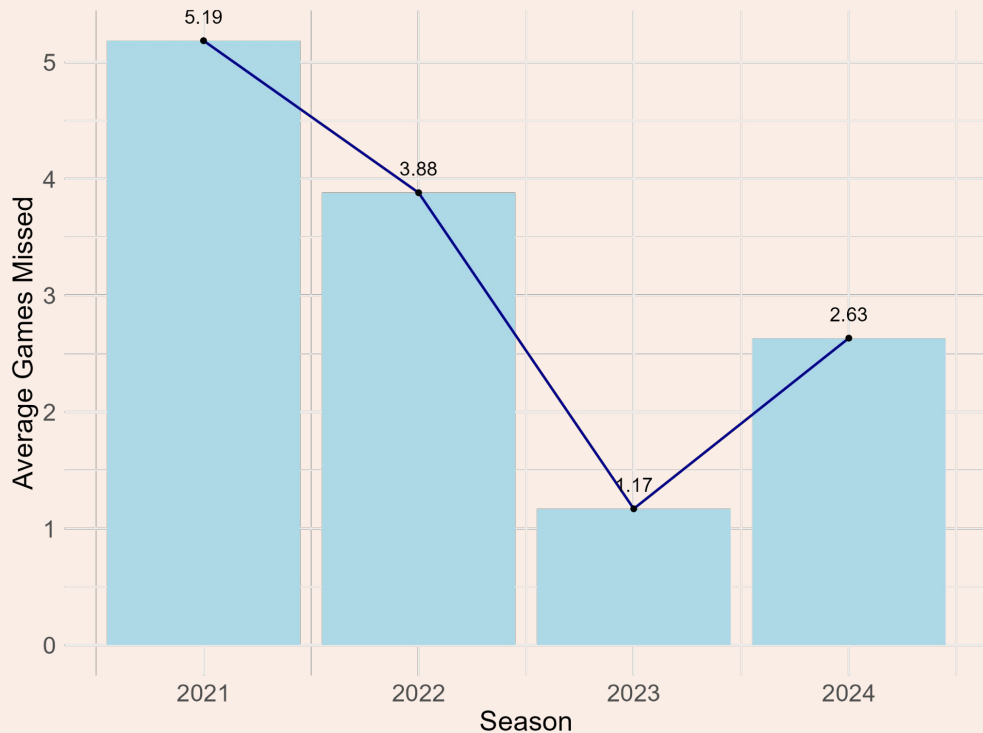
2024 had by far the lowest rate of injuries on non-return kickoffs.

Injury Rate on Returns



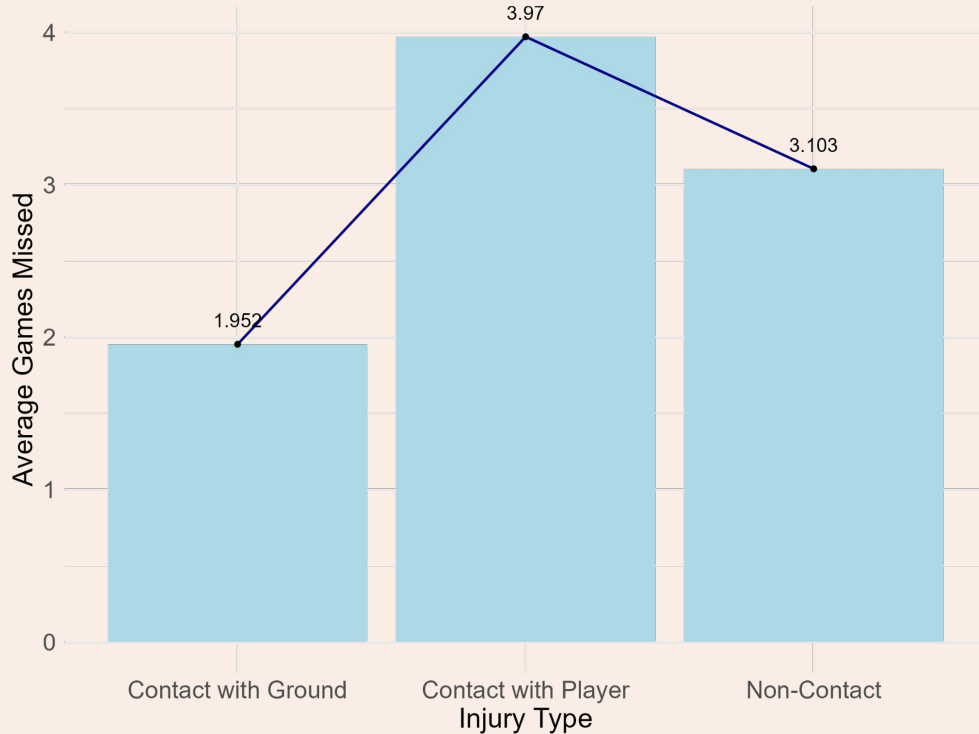
2024 had the lowest rate of injuries on returns

Average Games Missed Due to Injury by Season



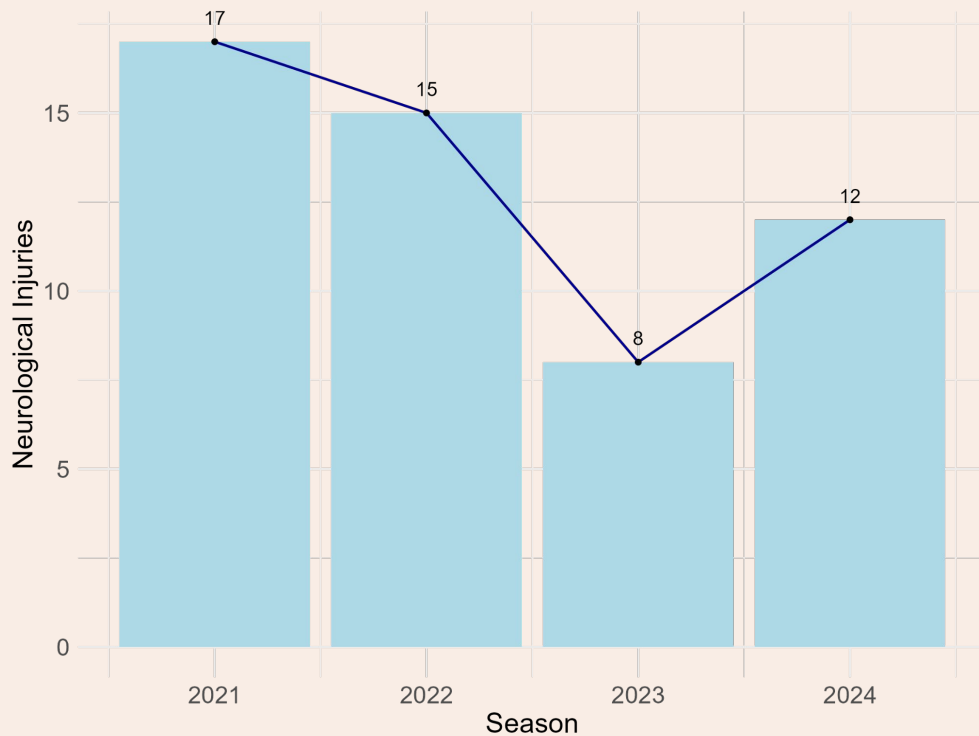
2023 is clear low-point in injury severity

Average Games Missed Due to Injury by Type



This is because 2023 had the highest rate of contact with ground injuries

Neurological Injuries by Season



2024 had slightly more than 2023, but performed better on a rate basis.



Statistical Analysis

Statistical Significance

After running various statistical tests, such as ANOVA and Kruskal-Wallis, we can determine that there is NOT a statistical significance that injuries vary and differ from year to year.

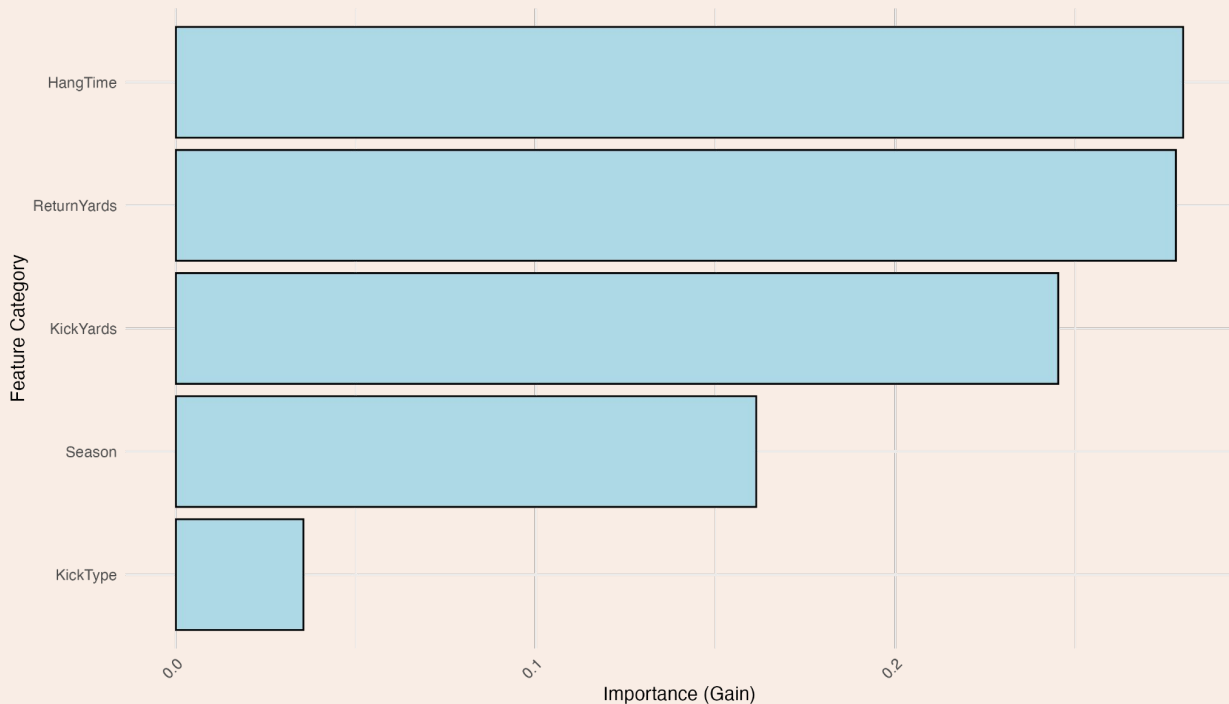




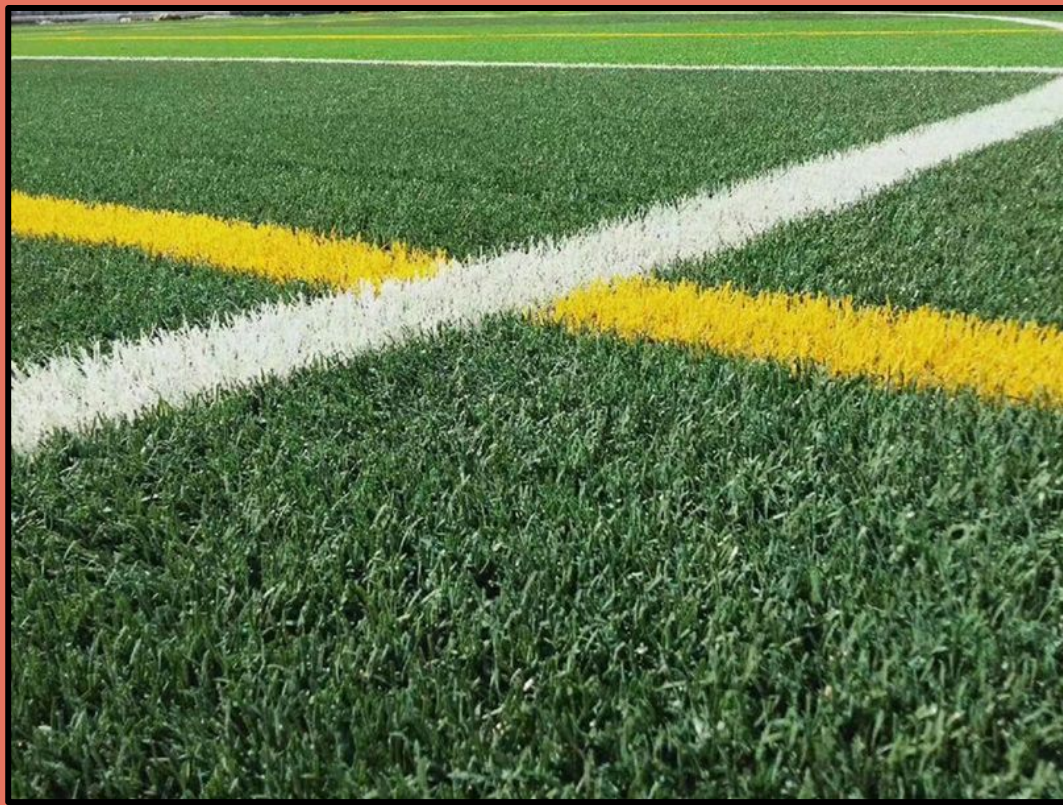
Predictive Modeling

Predicting Injuries/Severity

Top Predictors of Injury



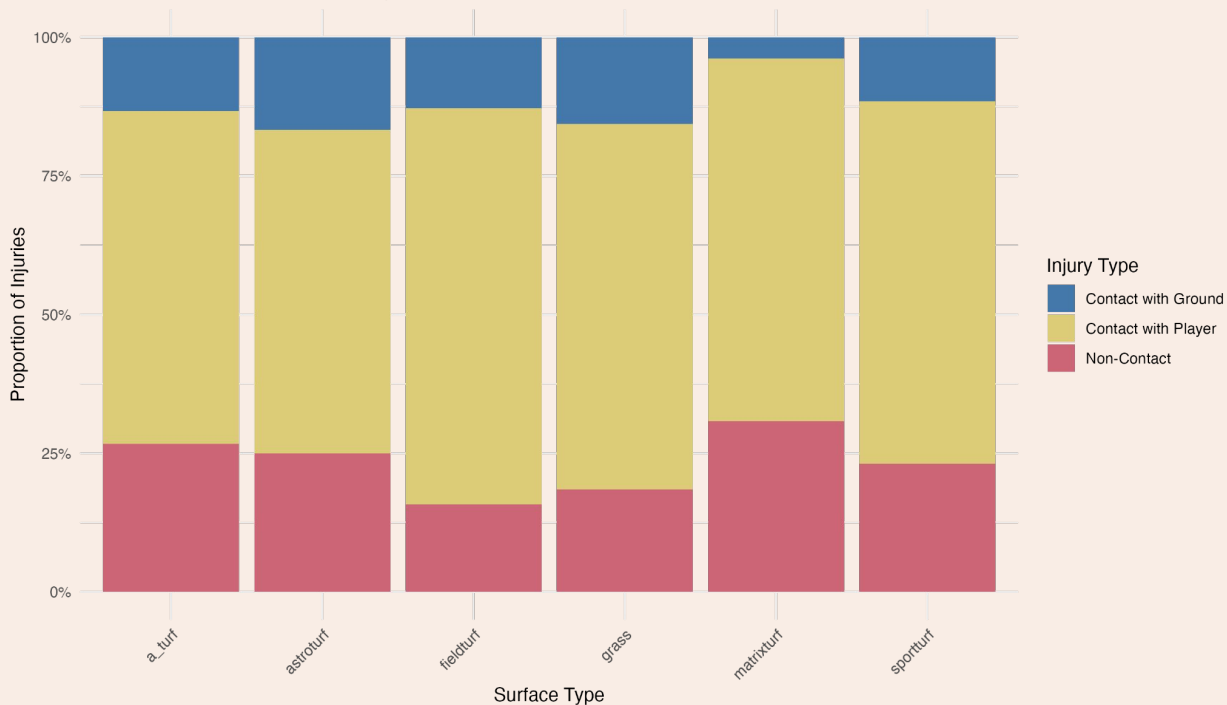
The response variable is a numerical representation of injury severity from 0 to 1. A 0 represents no injury.



Field Surfaces

Field Surfaces

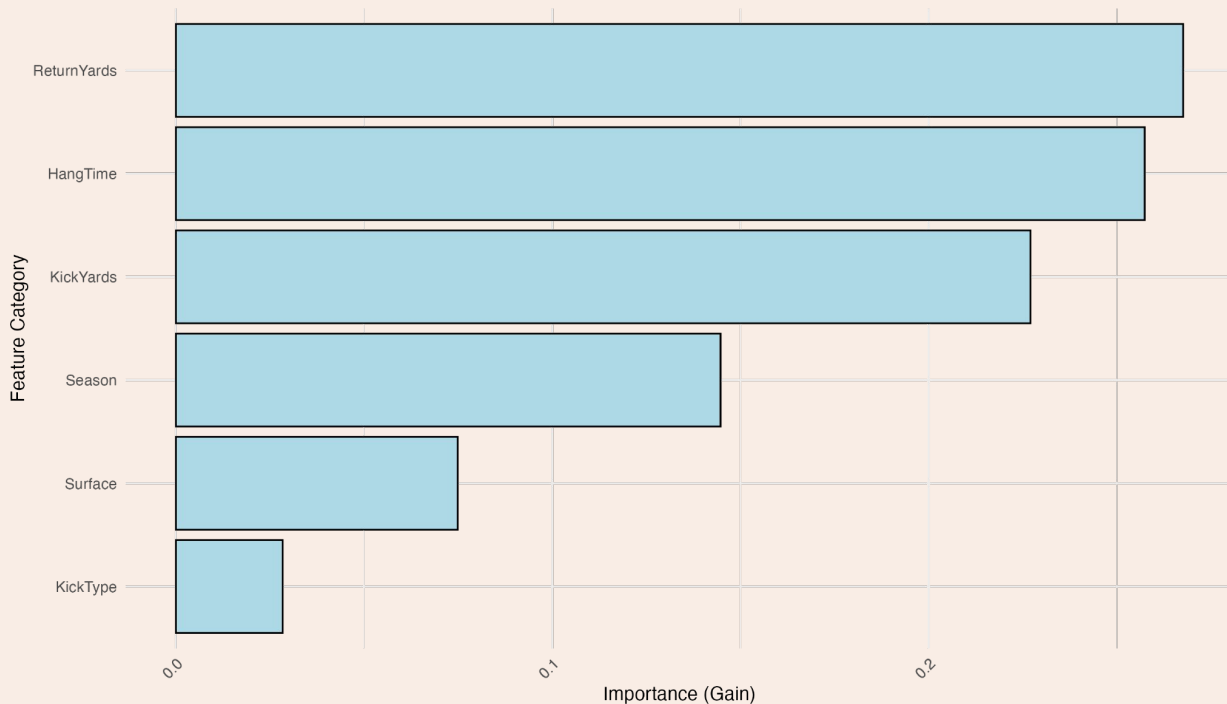
Standardized Breakdown of Injury Type by Surface



Mostly uniform, although
matrixturf greatly
reduced ground injuries.

Field Surfaces (Predictive Modeling)

Top Predictors of Injury (Aggregated Categories)



Surface Type is not a very strong predictor of injury or severity.

Conclusion

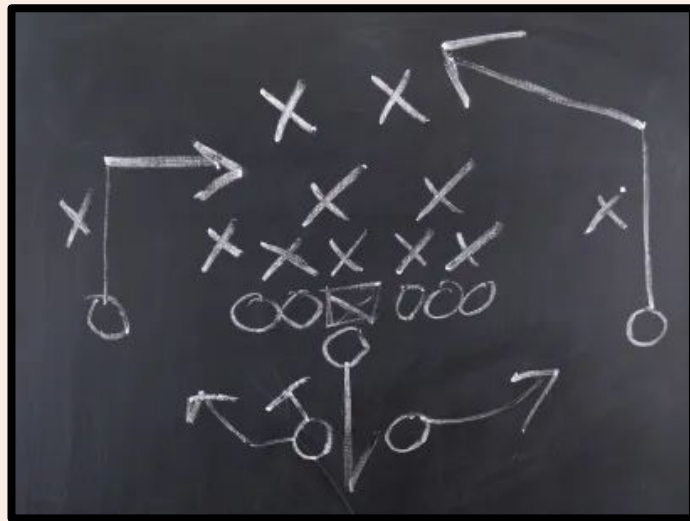




Our Advice

Optimal Kickoff Strategy

- 1** Teams should almost always kick into the end zone.
- 2** Teams should normally try to return if the ball is kicked into landing zone, but not if kicked into end zone.
- 3** If losing late in the game, returning is almost always the best option.



Rule Changes

Given that the injury decrease is not statistically significant, we advise the following:



- 1** The kick return is an exciting part of football, and so we recommend keeping it in the game.
- 2** The dynamic kickoff provided a great way to almost eliminate unnecessary non-return injuries.
- 3** Moving forward, we know that big returns and kicks with higher hang time cause more injuries, which prompts further examination.

Dynamic Kickoff Verdict

One year in, the new rules seem to have had their desired effect: improving game watchability while reducing injuries.

- 1** Touchback percentage is back in the 60% range.
- 2** Kick returns are helping boost scoring as returning teams have higher EPA than ever before.
- 3** Non-return injuries have been wiped out, and injuries are similar to 2023 levels even with far more returns.

THANK YOU

ANY QUESTIONS?



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